

VISUAL GOAL NAVIGATION FOR QUADRUPEL ROBOTS BASED ON ACTION DIFFUSION POLICY

Background and Motivation

Visual goal navigation is a core task in embodied intelligence that connects perception with decision-making. Given the current visual observation and a goal image, the policy must output executable local motion without relying on high-precision maps or global localization, thereby achieving autonomous movement from the current viewpoint to the goal viewpoint. As robots are increasingly deployed in unstructured open-world environments such as campus patrol, indoor guidance, last-mile delivery and home service, navigation can no longer be reduced to a pure geometric path-planning problem on a known map. It must instead handle weak priors, partial observability, dynamic obstacles and ambiguous goal specifications, which has motivated a paradigm shift from classical Simultaneous Localization and Mapping (SLAM)-plus-planner pipelines toward learning-based, image-conditioned policies that consume raw visual observations directly.

Recently, general navigation models such as General Navigation Model (GNM), Visual Navigation Transformer (ViNT), and Goal Masked Diffusion Policies for Navigation and Exploration (NoMaD) have introduced diffusion policies into visual navigation, modelling the future local trajectory as a conditional distribution and generating multi-modal candidates through iterative denoising sampling. This formulation has significantly stronger expressive power than regression-based end-to-end policies that collapse multi-modal decisions (such as turning left or right at a T-junction) into a single mean output. By modelling the full distribution of feasible local trajectories, diffusion-based policies are able to preserve multi-modality, sample diverse candidates, and integrate naturally with test-time selection and guidance mechanisms. However, closed-loop validation of these models has so far concentrated on wheeled platforms (LoCoBot, TurtleBot, Jackal), while deployment pipelines targeting quadruped robots remain underexplored. Quadruped platforms differ substantially from wheeled ones in gait-induced jitter, control-loop frequency, body-posture coupling, foot-impact disturbance, and real-time safety requirements. These differences pose new challenges for the collaboration between high-level diffusion policies and mid-/low-level control

interfaces, and they cannot be addressed by simply re-routing the wheeled-robot velocity command to a quadruped’s locomotion controller.

Objectives and Approach

This thesis takes NoMaD as the high-level navigation core and the DeepRobotics Lite3 quadruped as the target deployment platform, and addresses three deployment-oriented questions: (i) how the visual encoder, distance-prediction head, and diffusion policy head are coupled in the NoMaD implementation and where inference-level optimizations are possible without retraining the model; (ii) whether, without retraining, sampling acceleration via Denoising Diffusion Implicit Model (DDIM), conditional guidance via Classifier-Free Guidance (CFG), test-time search via Test-Time Scaling (TTS), and visual encoder replacement can meaningfully improve policy usability for resource-constrained edge deployment; and (iii) whether these optimizations can be integrated into a hierarchical closed-loop system for Lite3 and validated through Multi-Joint dynamics with Contact (MuJoCo) simulation as well as real-world deployment on the Jetson Orin onboard computer. The technical roadmap proceeds through unified-inference-module construction and optimization, system-level integration, and deployment-level validation, corresponding respectively to Chapters 2, 3, and 4 of the thesis. A deliberate emphasis is placed on bridging the offline-evaluation versus closed-loop deployment gap, ensuring that every algorithmic conclusion is re-examined under realistic timing, safety, and platform constraints.

Main Contributions

Unified inference module. A unified offline statistical pipeline is built to compare DDIM inference acceleration, CFG enhancement, Test-Time Scaling (TTS) search, and visual encoder upgrades under a single evaluation protocol. Concretely, DDIM-2 attains about $4.84\times$ speedup over the Denoising Diffusion Probabilistic Model (DDPM)-10 baseline on the GoStanford evaluation set without notable trajectory-quality degradation, with the mean-squared deviation against the baseline trajectory distribution remaining below 0.022. The CFG analysis, derived from score-function modification, establishes a guidance-scale selection rule and shows that CFG roughly doubles the per-step compute while delivering only marginal Forward-Progress gain in the diffusion-policy regime; hence CFG is best treated as a tunable knob rather than a default-on component. The TTS analysis decomposes total

gain into a nearly zero-latency *scoring-based selection* component (which re-ranks the eight candidates already produced by a single diffusion batch) and a linearly priced but rapidly diminishing *candidate-pool expansion* component (which spawns extra batches to enlarge the pool). TTS-8 wins the three-factor joint ranking at minimal cost, providing a novel interpretation of inference-budget allocation as “re-rank cheaply first, expand only when necessary.” To make the ranking criterion explicit, the configuration-level score used in the TTS tables is defined as $S_{\text{rank}} = \overline{\text{FP}} - 0.050 \overline{\text{Latency}} - 0.350 \overline{\text{LA}} - 0.150 \overline{\text{Smooth}}$, where latency is measured in milliseconds, lateral deviation and smoothness are penalized, and smoothness acts mainly as a weak regularizer because it is computed from the mean second-order trajectory difference. Encoder upgrades use a two-stage framework to compare EfficientNet-B0, ResNet-50, DINOv2-Small, and ConvNeXt-Tiny on identical offline cases; the encoder $\times$ DDIM / CFG / TTS joint ablation establishes that DDIM-2 and CFG default-off remain robust across all four encoders, while TTS gain scales with the initial candidate diversity of the chosen encoder.

System layer. A hierarchical closed-loop system—comprising a unified inference module, an explicit finite-state machine, a navigation host, a mid-level navigation-point-to-velocity mapping, and a platform interface—supports both MuJoCo and real-robot backends through one set of high-level code. The state machine (idle / stand / navigate / explore / recovery / estop / completed / failed) is decoupled from the physical platform via a uniform interface, so that switching between MuJoCo and Lite3 only changes the backend implementation rather than the high-level logic. The navigation host supports both batched JavaScript Object Notation (JSON) plans and interactive single commands, allowing capture, topological map building, navigation, and exploration to be composed within a single experiment session. Mid-level mapping converts the diffusion policy’s local navigation points into bounded forward, lateral, and yaw velocity commands, respecting Lite3-specific safety envelopes ($v_x \leq 0.4$ m/s in simulation, $v_x \leq 0.12$ m/s during real-robot deployment). In MuJoCo closed-loop experiments, Lite3 physically reaches the target in point-navigation and multi-goal tasks under a 0.5 m Euclidean-distance threshold, while the exploration mode verifies that the goal-masked, target-free inference chain remains executable on the same system. The same unified inference module that handles the four offline optimizations is re-used at deployment time, demonstrating that the algorithm-layer

findings translate directly into operational system behaviour.

Deployment layer. For the Jetson Orin onboard host and the Lite3 upper-computer link, the thesis develops a staged commissioning procedure, multi-level safety mechanisms, and an image-recording convention. Stage 1 (Orin standalone commissioning) validates camera capture, model loading, inference benchmarking, and the end-to-end pipeline in isolation from the robot, so that any inference-side defect is discovered before the platform is energized. Stage 2 (Orin and Lite3 joint commissioning) progresses from network and heartbeat verification, through standup and low-speed twist tests, to interactive and goal-directed navigation under conservative speed limits. The bridging layer wraps the Lite3 Software Development Kit (SDK) with a 25 Hz velocity-refresh thread and a 5 Hz heartbeat thread, repeating the most recent clipped velocity between higher-level NoMaD inference cycles to guarantee continuous control flow even when the policy runs at single-digit hertz. Multi-level safety guards include Not a Number/infinity (NaN/Inf) clipping, magnitude limits, fallen-robot detection, and a connection-stale threshold of two seconds; any inference anomaly or platform fault is routed to a soft emergency stop. Real-robot experiments on multiple indoor scenes with pre-collected topological maps record closed-loop frequencies of 2.89 Hz (DDPM baseline), 6.80 Hz (DDIM-2 + CFG=0 + TTS-8), 3.43 Hz (DDIM-2 + CFG=2 + TTS-8), and 5.90 Hz (DDIM-3 + CFG=0, no TTS), with a 100% success rate across all four configurations under the present conservative speed setting. Unlike MuJoCo, the real robot does not use global ground-truth coordinates for success evaluation; success is determined by a visual-topological arrival rule in which the selected topological node reaches the goal node and the NoMaD distance head predicts a distance below the dimensionless threshold $\tau_{\text{arrive}} = 3.0$, with no emergency stop or manual intervention. The frequency ranking is fully consistent with the offline analysis in Chapter 2: DDIM step count dominates latency, CFG approximately halves the achievable rate, and TTS-8’s overhead is negligible because re-ranking eight already-generated candidates does not require an extra diffusion batch.

Experiments and Conclusions

Offline results indicate that DDIM-2 plus optional CFG and configurable TTS-8 is the most promising configuration to bring into closed-loop validation, and the encoder upgrade pipeline supplies a reusable framework for future visual-backbone exploration rather than a single best-encoder verdict. MuJoCo experiments confirm that the hierarchical system

can robustly organize quadruped navigation, multi-goal execution, exploration, recovery, and emergency-stop transitions through one shared control architecture. Stage-1 validation on Jetson Orin demonstrates the baseline feasibility of the real-robot pipeline (about 7 Hz benchmarked NoMaD inference, $\approx 151\text{--}157$ ms steady-state pipeline cycle once warm-up cost is amortized), and Stage-2 multi-scene results provide the first real-robot evidence that diffusion-policy-based image-goal navigation can run end-to-end on a Lite3 quadruped at single-digit hertz while still completing every commissioned trial. Crucially, the fact that even the slow 2.89 Hz baseline achieves 100% success demonstrates that the closed-loop system has substantial robustness margin: the conservative 0.12 m/s speed cap means each cycle’s displacement is only about 4 cm—far smaller than the topological map node spacing—so high-level latency tolerance is absorbed by the topological map sliding window and the 25 Hz bridge replay rather than manifesting as catastrophic drift.

Limitations and Future Work

Three limitations are explicitly acknowledged. First, the encoder upgrade conclusions remain offline statistics plus simulator validation, and a final real-robot-conditioned encoder ranking will require benchmarking inference latency, image-distribution robustness, and failure modes directly on the Orin/Lite3 stack with real cameras. Second, the present real-robot statistics, while sufficient to show that all four inference configurations succeed under the current scenario scale, do not yet include large-sample success-rate confidence intervals or formal trajectory-deviation distributions; expanding the experiment matrix to outdoor scenes, dynamic obstacles, and longer corridors should enhance differentiation between configurations and unlock the latency margin that the fastest scheme has reserved. Third, the topological map is still collected offline by an operator; an online maintenance loop that refreshes nodes when the environment changes would improve long-term autonomy. Future work will therefore focus on (i) finalizing the on-robot encoder choice with end-to-end real-world metrics, (ii) running large-scale repeated trials with formal statistics, (iii) generalizing the unified inference module and state machine to additional platforms such as the Tron1 humanoid robot to test cross-embodiment portability, and (iv) extending image-goal conditioning to language-grounded targets, allowing operators to issue natural-language commands such as “go to the red door at the end of the corridor” that get translated into NoMaD-compatible visual conditions.

The thesis extends NoMaD from a wheeled-platform research framework into an analyzable, simulatable, and deployable research pipeline for quadruped platforms, distilling a reusable inference-optimization protocol, a hierarchical closed-loop system organization, and a real-robot safety-guard checklist. Together, these artefacts provide a practical basis for subsequent research on diffusion-policy-based visual navigation for quadruped robots, and they lower the engineering threshold for follow-up work that wishes to validate algorithmic ideas under genuine on-robot conditions rather than only on offline benchmarks.